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□□□□ A □□□ Tick □□□500ms □□□□□□□□□□□□□□□□□□□□

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1.1 □□□□

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[illegible]

2.1 [illegible]

[illegible]A/B/C/D/E[illegible] 5 [illegible] 27,601 [illegible] Tick [illegible] 33 [illegible]

[illegible]

[illegible] E [illegible]

$$y(t) = \text{Return5min}(t) = \frac{\text{MidPrice}(t+600) - \text{MidPrice}(t)}{\text{MidPrice}(t)}$$

[illegible] $\text{MidPrice}(t) = \left(p_1^{b(t)} + p_1^{a(t)}\right)/2$[illegible]600 [illegible] Tick ≈ 5 [illegible]

[illegible] $\hat{y}(t) = f_\theta(\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \boldsymbol{x}_t)$ [illegible] t [illegible] t [illegible]

[illegible]

- [illegible]IC[illegible]$\text{IC}_d = \text{Pearson}(\hat{\boldsymbol{y}}_d, \boldsymbol{y}_d)$[illegible][6][illegible] 53 [illegible]
- ICIR[illegible]$\text{ICIR} = \overline{\text{IC}}/\sigma(\text{IC})$[illegible] IC [illegible]ICIR [illegible] IC [illegible][6][illegible] 55 [illegible]

2.2 [illegible]

2.2.1 [illegible]OVI[illegible]

[illegible]OVI[illegible] IC≈0.26[illegible] Kyle [illegible]

$$\text{OVI}(t) = \frac{\sum_{i=1}^{5} V_i^{b(t)} - \sum_{i=1}^{5} V_i^{a(t)}}{\sum_{i=1}^{5} V_i^{b(t)} + \sum_{i=1}^{5} V_i^{a(t)} + \varepsilon} \in [-1, 1]$$

[illegible] $\varepsilon > 0$ [illegible]V_i^b, V_i^a [illegible] i [illegible]

“OVI [illegible]”[illegible]Pulse[illegible] OVI [illegible] SMA [illegible]

$$\text{OVI_pk}(t) = \overline{\text{OVI}}_{k(t)} - \overline{\text{OVI}}_{600}(t), \quad k \in \{15, 30, 60\}$$

[illegible] $\overline{\text{OVI}}_{k(t)} = \frac{1}{k}\sum_{s=t-k+1}^{t} \text{OVI}(s)$ [illegible] k [illegible] Tick [illegible] $k < 600$ [illegible]$\text{OVI_pk} > 0$ [illegible]

2.2.2 [illegible]TI[illegible]

[illegible]

$$\text{TI}(t) = \frac{V_{\text{buy}(t)} - V_{\text{sell}(t)}}{V_{\text{buy}(t)} + V_{\text{sell}(t)} + \varepsilon}$$

[illegible] $V_{\text{buy}}, V_{\text{sell}}$ [illegible]TI [illegible] Kyle [illegible] u_t[illegible]

[illegible]

2.2.3 □□□□□□□□□OBI□

□□ Glosten-Milgrom □□□□□□□□□□2−5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

$$\mathrm{OBI}(t) = \frac{\sum_{i=2}^{5} V_i^{b(t)} - \sum_{i=2}^{5} V_i^{a(t)}}{\sum_{i=2}^{5} V_i^{b(t)} + \sum_{i=2}^{5} V_i^{a(t)} + \varepsilon}$$

□□ OVI□1−5 □□□□□□OBI □□□□□“□□□□□”□□□□□□□□□□□ OVI □□□□□□□□□□□□

2.2.4 AMH □□□□□□□□□□□

□□□□□□□□□AMH□[5] □□□□□□□□□□□□□□□ D1 □□□□□□□□□□□ D4 □□□□□□□□□□□□□□□□□□□□

[redacted] i [redacted] $[t-W, t-d)$ [redacted] IC [redacted]

$$\mathrm{IC}_{i(t)} = \mathrm{Pearson}(\left\{\hat{y}_{i(s)}\right\}_{s=t-W}^{t-d},$$

$$\{y(s)\}_{s=t-W}^{t-d})$$

[redacted] $W = 600$[redacted] 5 [redacted]$d = 600$[redacted] $y(t-d) = \mathrm{Return5min}(t-d)$ [redacted] t [redacted]

EWMA [redacted] IC [redacted] $T_{\frac{1}{2}} = -\ln \frac{2}{\ln(1-\beta)} \approx 99$ [redacted]$\beta = 0.007$[redacted]

$$\widetilde{\mathrm{IC}}_{i(t)} = (1-\beta) \cdot \widetilde{\mathrm{IC}}_{i(t-\Delta)} + \beta \cdot \mathrm{IC}_{i(t)}$$

[redacted] $\Delta = 15$[redacted] 15 [redacted] Tick [redacted] Softmax [redacted]

$$w_{i(t)} = \frac{\exp\left(\widetilde{\mathrm{IC}}_{i(t)} \cdot \tau\right)}{\sum_{j=1}^{N} \exp\left(\widetilde{\mathrm{IC}}_{j(t)} \cdot \tau\right)}, \quad \tau = 17$$

[redacted] τ [redacted]τ [redacted] IC [redacted]

$$\hat{y}(t) = \sum_{i=1}^{N} w_{i(t)} \cdot \hat{y}_{i(t)}$$

[redacted]

- [redacted]MTC[redacted] 1 [redacted] stable_prior $= 2.0$ [redacted]
- Niche [redacted] 27 [redacted] 0[redacted] IC [redacted] IC [redacted]

[redacted]

3.1 [redacted]

3.1.1 [redacted]

[redacted] E [redacted]

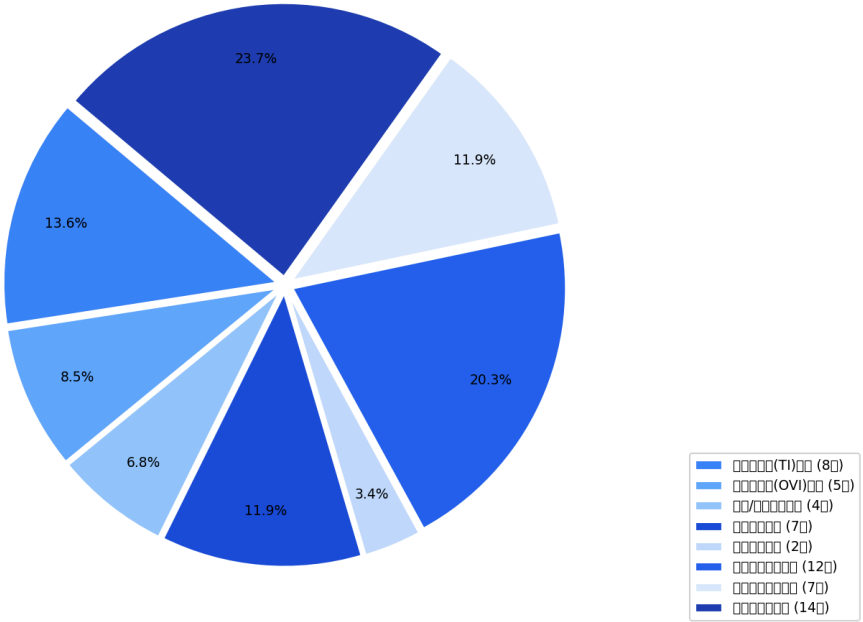
$$y(t) = \frac{\mathrm{MidPrice}(t+600) - \mathrm{MidPrice}(t)}{\mathrm{MidPrice}(t)}$$

$t+600$ [redacted] t [redacted] t [redacted] $\hat{y}(t)$[redacted] $y(t)$ [redacted] IC [redacted]

3.1.2 [redacted]

[redacted](1) [redacted]——[redacted] t [redacted](2) [redacted] IC [redacted]——[redacted] 5 [redacted] IC [redacted](3) [redacted]——[redacted] < 0.7[redacted](4) [redacted]——[redacted]

3.1.3 [redacted]59 [redacted]8 [redacted]



□ 2　□□□□□□□□□□□ 59 □□□□8 □□□

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(1) □□□□□□TI□□□□8 □□□ $\mathrm{TI_p}k = \overline{\mathrm{TI}}_k - \overline{\mathrm{TI}}_{600}$□$k \in \{15, 30, 40, 60\}$□□□□□□□□□□□□□□□□□□□□ EMA □□ $\mathrm{TI_ep}k$□

(2) □□□□□□OVI□□□□5 □□□ □□ 2.2.1 □□□□□□ IC≈0.26□□□□□□□□

(3) □□/□□□□□□□4 □□□ □□□□□□□□□□ ONI□Order Number Imbalance□□ TNI□Trade Number Imbalance□□□ OVI □□□□□□□□□□□□

(4) □□□□□□□□7 □□□ A/B/C/D □□□□□□□□□□□□□□□□□□ OBI□□□ TI □□□□□ OVI □□□□□□□□□□□□□□□□□□

(5) □□□□□□□□2 □□□ `aft_13800`□□□□□□□$t \geq 13800$s□□ `aft_12000`□□□□□□□□□□□□□

(6) □□□□□□□□□□12 □□□ □□ E □□□□□□ 5 □□□□□□□□□□□□□□□□□□□□ E □ 30/60/120/300/600/900 Tick □ □□□□□□

(7) □□□□□□□□□□7 □□□ □□□□□□□□□OBI□2–5 □□□□□□□□□□□□□□□□□□□□

- `_RunEMA(span)`□□□□□□□□□$\mathrm{EMA}_t = (1-\alpha)\cdot \mathrm{EMA}_{\{t-1\}} + \alpha \cdot x_t$□$\alpha = \frac{2}{\text{span}+1}$□

3.2.2 □□□□□

□□□ k □ Tick□k □□□□□□□□□□

$$\min_{\boldsymbol{w},b}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{w}-b\|_2^2+\alpha\ \|\boldsymbol{w}\|_2^2$$

□□□□□□$\hat{\boldsymbol{w}}=\left(\boldsymbol{X}^\top\boldsymbol{X}+\alpha\boldsymbol{I}\right)^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

Ridge □□ OLS □□□□□□□(1) □□□□□□□□□□□ OVI/TI □□□□□□□□□□OLS □□□□□□□□□□Ridge □□ α □□□□□-□□□□□□(2) □□ LightGBM/□□□□□□Ridge □□□□□□□ $n=138000$ □□□□□□□ $p=59$ □□□□□□□□□□□□□□□□□□□ α □□□□□□□□□□□□□□ $\alpha=150$□□□□□□□□□OVI □□□□□ $\alpha=10$□□□□□□□□□□□□□□□□□□

- Iter3–8□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□IC □□□ 0.279□
- Iter9–12□□□ 0.30□□ □□□□□□□□□□□ 5 □□□□□□□□ IXN niche □□□□IC □□□□ 0.30□□□ 0.300□
- Iter13–19□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□ OVI □□□□□□IC □□□□□ 0.313□ICIR □ 7.82□
- Iter20□□□□□□□□□□□□ LOO □□□□ 54 □□□□□□ 28 □□□□□ IC □□□ 0.330□ICIR □ 7.82 □□□ 12.59□+61%□□ □□□□□□□□□□□□□□□ IC □□□□□

3.4.4 □□□□□□

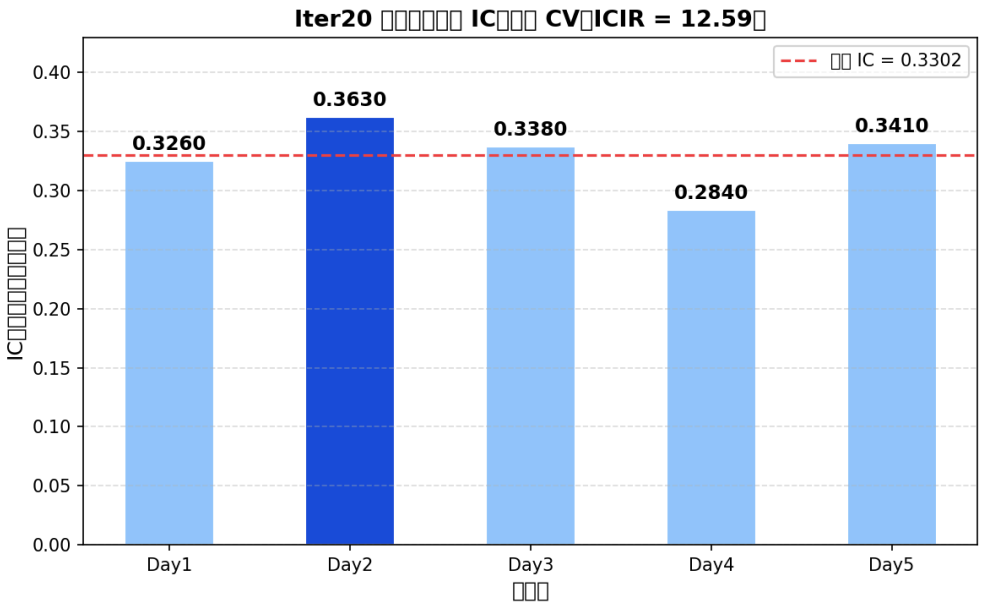
□□	□	□□
ENSEMBLE_WINDOW	600	□□ IC □□□□□□ 5 □□□
ENSEMBLE_TEMP	17	Softmax □□□□□□□□□□
ENSEMBLE_FLOOR	0	□□□□□□□□□□□ 0□
RETURN_DELAY	600	IC □□□□□□□□□Tick□
ENSEMBLE_UPDATE_FREQ	15	□□□□□□□□ 15 Tick□
ENSEMBLE_EWMA_BETA	0.007	IC □□□□□EWMA□□□□□ 99 Tick□
STABLE_PRIOR	2.0	□□□□□□□□□□□□□
NICHE_INIT_WEIGHT	0	niche □□□□□□□□□

□ 2　□□□□□□□□

3.5　□□□□□□□□□□□

3.5.1　□□ IC □□

□ 3 □□□□□□□□Iter20□□□□ CV □□ IC□

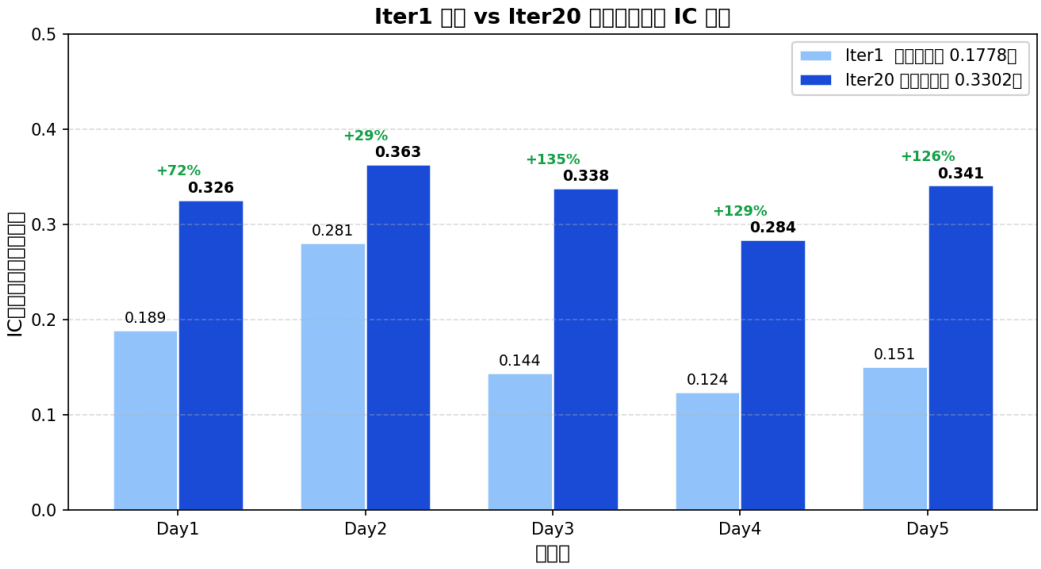


□ 4　Iter20 □□□□□□□ CV □□ IC□□□=0.330□ICIR=12.59□

□□□□ IC □□□□□Day1=0.326□Day2=0.363□Day3=0.338□Day4=0.284□Day5=0.341□□□ IC=0.3302□□□□=0.026□ICIR=12.59□Day4□□□□□□□□IC □□□0.284□□□□ AMH □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ IC □□□□ 0.026□□□□□□□□□□□□□□

3.5.2 □□□□

□ 4 □□□□ Iter1□6 □□□□□□□□ Iter20□28 □□□□□□□□□□□ IC □□□

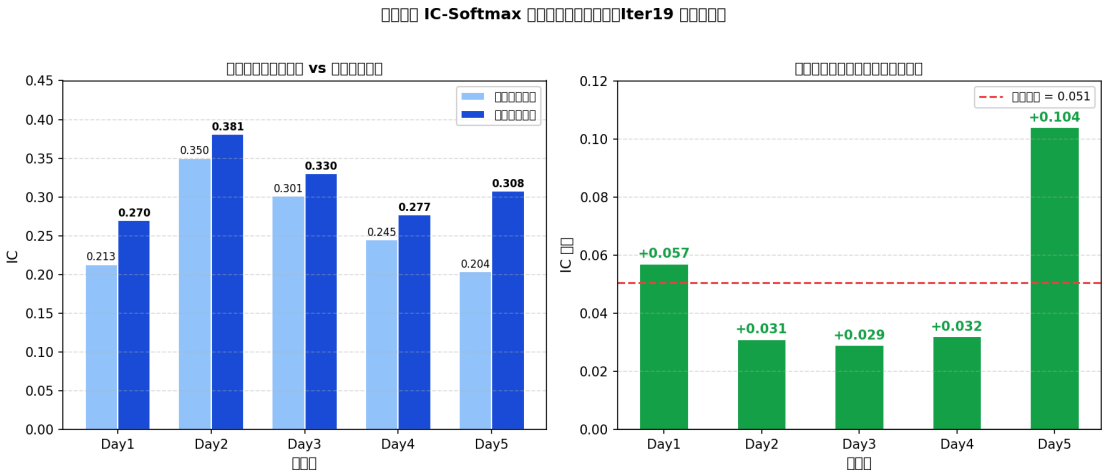


□ 5 Iter1 □□ vs Iter20 □□□□□□□ IC □□

□□□□□Iter20 □□□□□□□□□□□□□□□Day1 □□ 72.5%□0.189→0.326□□Day3 □□ 134.7%□0.144→0.338□□Day4 □□ 129.0%□0.124→0.284□——□□□□□□Day1/3/4□□

3.5.3 □□□□□□□□

□ 5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□



□ 6 □□□□□ vs □□□□□□□□ IC □□□□□□□□□Iter19 □□□□□□

□□□□□□□□ 5 □□□□□□□□□□□□□□□□+0.051□□□□□□ IC-Softmax □□□□□□□[10]□□□□□□□□ Day5□+0.104□□□□ niche □□□□□□ IC □□□□□□□□□□□□□□□□□□□□□□□□

3.5.4 □□□□□□

□□□□	□□□	□□
□□ CV □□ IC	0.3302	□□□□□□□out-of-sample □□
□□ CV IC □□□	0.026	□□□□□□□□□□
ICIR□IC/Std□	12.59	□□□□□ IC□□□□□
Penalized□M−0.5S□	0.317	□□□□□□□□□□□□□
□□ 4 □ IC	0.311	□□□□□□□□□□□□□□
□-□ gap	−0.019	□□=□□□□□□□□□□□□
□□□□□vs □□□	+0.051	□□□□□□□□□□□□
□□□□□	28 □□□□59 □□	□□ Iter19□54×59□□□□

□ 3 □□□□□Iter20□□□□□□□

□□□□□□□

4.1 Leave-One-Model-Out□LOO□□□□□□□□□

□□□□□□□□□□□□“□□□□”□□□□□□□5 □□□□□□□□□□□□□□

4.4 [illegible]

[illegible] 59 [illegible] $O(1)$ [illegible] Tick [illegible] SMA [illegible] EMA [illegible][12] [illegible]

[illegible]

5.1 [illegible] Alpha [illegible]

[illegible] Alpha [illegible] $\hat{y}(t)$ [illegible] $\hat{y}(t) > \theta$ [illegible] $\hat{y}(t) < -\theta$ [illegible] θ [illegible] CV IC=0.330 [illegible]

5.2 [illegible]

[illegible]

5.3 [illegible]

[illegible]

- [illegible]
- [illegible] ETF [illegible]
- [illegible] Ridge [illegible] Online Learning [illegible]

5.4 [illegible]

28 [illegible] OVI [illegible] niche [illegible]

[illegible]

[illegible] Tick [illegible]

(1) [illegible] Kyle（1985）[illegible] Glosten-Milgrom（1985）[illegible] OVI、TI、OBI [illegible] AMH [illegible]

(2) [illegible] LOO [illegible] 54 [illegible] 28、IC [illegible] 0.313 [illegible] 0.330、ICIR [illegible] 7.82 [illegible] 12.59 [illegible] +61%。[illegible]

(3) □□□□-Niche □□□□□□□□□□□ OVI □□□“e_sect_lag_gap”□IC=0.203□□□□□□□□□□

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